② Static Website: (simple)

↳ Trying to set up LAMP stack

L → Linux
A → Apache
M → Mysequal
P → PHP scripting language/perl/python.

↳ static website Hosting on Git ④ in link-to-Git

step ① Apache setup [Installing Dependencies and requirements]:

→ cd ~ or cd /home/msis or cd

→ sudo apt-get install apache2

→ whereis apache2

Validation if apache is installed or not:

① ⇒ sudo systemctl status apache2

to see the status of the Service is executing or not

② ⇒ web Browser → localhost:80 or http://172.18.187.36

My system ip address

Note::

systemctl ⇒ sudo systemctl status start ______ Name of service

sudo systemctle " stop ______

sudo " " restart ______

On restarting the service will restart automatically ⇒ sudo " " enable ______

Wont restart automatically ⇒ sudo " " disable ______ step② : Configuration of services (just checking)

→ sudo nano /etc/apache2/sites-available/000-default.com

<VirtualHost * :80>

listen on all the server in case a http packet is sent on 80 port.

000-default.conf whenever will render the data into website mentioned

Default root file of all the Web server

step③ :

→ cd /var/www/html

→ ls

index.html is have Create new html folder under this path

→ Download a file from Browser

→ html5up-paradigm-shift.zip

→ cd

→ mkdir myapp

→ ls

→ cd myapp

→ sudo cp /home/msis/Downloads/html5up-*.zip .

Copy

Indicating current path

→ ls

→ sudo apt-get install unzip

→ sudo unzip html5up-paradigm-shift.zip

→ ls

recursively

→ sudo rm -rf html5up-par*.zip

→ ls

→ cd ..

→ sudo cp ~~rf~~ -rf myapp/* /var/www/html/.

→ ls /var/www/html/

Now we have to eddit the 000-default.conf file
Eddit by adding /html → /html/myapp

Note:

mv → is cut and paste or rename file

→ Now restart the appache 2 after changes

→ sudo systemctl restart apache2

→ go to ~~the~~ webbrowser & reload the page

→

Assignment: Try cafe-static website ①

↳ Cloan from git and then make above changes

→ sudo git clone https://github.com/Sreem/Cafe-static-web.git

→ sudo a2ensite myapp.conf

→ Sudo a2ensite Cafe-static-website.conf

→ cd /var/www/html.

→ /var/www/html$ ls -

→ cd /etc/apache2/sites-available

→ ls -

→ sudo nano myapp.conf ← add 2 myapp.conf - Cafe static-website. make change conf add

→ sudo nano /etc/hosts ← domain name resolution. myapp.conf static website

→ 172.18.182.22 music.cse.com.

172.18.182.22 myapp.com.

172.18.182.22 mycafe.com -

(to disable a site):

→ sudo a2dissite 000-default.conf :

→ cd .

→ sudo a2ensite 000-default.com .

→ sudo a2ensite myapp.conf .

→ sudo a2ensite Cafe-static-website.conf .

→ check on Chrome myapp.com & ~~myapp static-website~~ website

Connection string: Clambstack (Frontend - System - Database)

→ sudo git clone https://git. . . /Cafe-Dynamic-website.

CRE → Create, Remove, Read, Execution

~~Sed Sedd~~

~~Setting up a~~

Setting up a lampstack and Cafe-dynamic-web:

→ sudo git clone https://github.com ... .git

→ sudo apt-get install php

→ sudo apt-get install ~~php~~ php php-mysql
mysql-server
mysql-client
libapache2-mod-php

Interface b/w php & Apache2 → libapache2-mod-php

$\underbrace{\text{php php-mysql mysql-server mysql-client libapache2-mod-php}}_{\text{Lamp stack}}$

Note: Mysqld → pid → 14347.

→ sudo cp -rf mompopcafe/ /www/html/.

→ ls /var/www/html/

→ cd /etc/apache2/sites-available/

→ ls

→ ~~sudo nano mom~~

→ sudo cp 000-default.conf mompopcafe.com

→ sudo nano mompopcafe.com

Edit the
ServerName ~~xxx~~ mompopcafe.com

DocumentRoot /var/www/html/mompopcafe

→ cd

→ sudo nano /etc/hosts/ → Resolution, diff icladdl Bon the su

make changes by adding
172.18.182.36 mompopcafe.com

My Local IP address Number

→ sudo a2ensite mompopcafe.com

→ sudo systemctl ~~sol~~ restart apache2

- sudo a2dissite 000-default.com

Sql db creation steps: (root-3306)

→ cd

→ sudo mysql -u root -p ⏎

Note!

-P3307 -h172.18 -p

Instead of other port   External IP

→ Create database cafedb.

→ Show databases

→ Create USER 'bda'@'Localhost' IDENTIFIED BY 'msis@123';

→ Grant all privileges on cafedb.* To 'bda'@'Localhost';

→ Flush Privileges;

→ Exit;

Check if Everything there
/Cafe-Dynamic-Website

⇒ ~~Cafe~~ sudo mysql -u bda -p
pass: msis@123

⇒ Show databases;
⇒ Use Cafedb;
⇒ Show tables;
⇒ Source Cafe_Dynamic_Website/mompopdb/Create-db.sql;
or
Source Create-db.sql
⇒ Show tables;
⇒ select * from order;
⇒ Exit
⇒ Config file (Connection string update)
⇒ cd /var/www/html/mompopcafe/
⇒ Sudo nano getAppParames.php
db_url = "localhost"
db_name = "Cafedb";
db_user = "bda";
db_password = "msis@123".

⇒ sudo systemctl restart apache2

Assignment : eCommerce.Com

⇓

→ sudo git clone https://...../phpmysql-app

Git (Version Control System) :-

↳ .git folder in our machine (Local Repo)

↳ Central Repo → on pushing it get .git folder saved

↳ git add → to push to staging area to track the files and update or else it will be untracked.

steps:

1. Loggin to git
2. Create a Repo → devop-Banking App

Note:

• gitignore → some files that I dont want to track. or want to ignore

③ Cmd:

echo } Check
cat } whats
tt } the diff

↳ mkdir myBankingapp

↳ cd myBankingapp/

↳ ls -la

↳ (In order to get git file) sudo git init

↳ ls -la { Now .git will be available in Working directory

↳ sudo git remote add origin $\underbrace{\text{https:....Banking-app.git}}_{\text{this will Origin Repo}}$

↳ sudo git pull origin main

↳ sudo git branch

↳ sudo git branch -r

↳ sudo git Branch -a ⇝ gives Local & Central Repo

↳ Sudo git Branch main ⇝ Create a Branch

↳ ls

↳ sudo git branch

↳ ls

↳ sudo git checkout main

↳ ls

↳ touch index.php

↳ ~~Cat~~ &

↳ Cat >> index.php ⇝ Display, append, redirect

Note: opens up a file write something

↳ Control C to get out

↳ Cat index.php

↳ Sudo git status

↳ ~~git add~~ sudo git add index.php to stage the changes

↳ Sudo git status

~~↳ sudo git add index.php~~

↳ ~~Sudo~~ Cat >> index.php

Updated Banking app

(>> → append
 > → Rewrite)

↳ Cat index.php

↳ sudo git status

↳ sudo git Commit index.php -m "first-version of my index page"

↳ ~~git~~ sudo git push origin main

↳ Profile → settings → dev settings → personal access tokens

→ token classic → generate new token classic

→ touch about.php Contact.php

→ ~~Gaa~~ ls

→ Cat ~~Gea~~ >> contact.php

→ Cat >> about.php

↳ Sudo git add. or Sudo git add -A
(all more than one Untracked file

MP4
MD5 } Red
SHA256.

↳ Sudo git Commit. or
Sudo git Commit -a -m "added 2 more files"

↳ Sudo git log → log of all changes.
↳ Sudo git log --oneline Commit ID → Md4, md5
cryptographic algo -40Bit.
SHA256.

~~↳ Sudo git log --on~~

↳ Sudo git log --online --graph

↳ Sudo git push origin main

(*) git Feature enginnering :- Branch changes.

↳ Sudo git Branch → *Main

↳ ls

↳ Sudo git checkout -b fd.

↳ Sudo git branch.

↳ ls → all the files in ~~same~~ main

(*) what changes done in *fd wont effect the main

↳ touch fd.php

↳ Cat >> fd.php

↳ Cat fd.php
↳ ls
↳ Sudo git add fd.php
↳ Sudo git Commit fd.php -m "Feature Called fd".
↳ sudo git checkout main
↳ ls . ⇝ Wont be (Reflected).
⊛ Merging with "merge" → 1 Method.
↳ Sudo git checkout main
↳ Sudo git merge fd .
↳ ls ⇝ file will be merged .
↳ sudo git push origin main
⊛ Merging with PR.
↳ Sudo git checkout -b rd .
↳ Sudo git branch .
↳ ls .
↳ touch rd.php
↳ Cat >> rd.php
↳ ls .
↳ Sudo git add rd.php
↳ Sudo git Commit rd.php -m "Rd page added".
↳ Sudo git push origin rd .
→ goto github (Create a pr)
→ Create a pull request

Jenkins installation:

↳ 172.18.182.22:800

↳ Intall Jenkins (from the file)

↳ Jenkins_install_steps.txt

↳ sudo apt-get purge --auto-remove jenkins

↳ Username → jenkins

↳ Pass → jenkins@123

↳ IP address → 172.18.182.36/24

↳ 1st github repo to run simple Java maven based console

↳ Simple-Java-maven-app (github repo)

↳ Jenkins → new item → JavaCompile → Freestyle project → ok

↳ Source Code Management → git → paste git URL of Repo

↳ Branch specifier → */Master

↳ Triggers → manual

↳ Build steps

↳ invoke top-level Maven targets

↳ setting → tools → maven → mvn-maven → save app

↳ Project → ~~Config~~ ~~Jenkins~~ Java-Compile → Config → invoke top level Maven targets

↓

mvn-maven → Compile → Save

[Life cycle goal of Maven Note: Compile validate test verify install package ...]

→ Build Now → Console output → Success.

→ New item → Java - CodeReview → Free style.
↓
Copy from
↓
Java-Compile
↓
OK

→ Invoke top-level Maven targets
↓
maven
goals → validate -P metrics pmd:pmd checkstyle:checkstyle findbugs:findbugs

→ git → paste URL

→ Branches → */master

→ Save → Build Now → Console output.

Publishing Result:

① Plugin Warnings

→ settings → plugins → Available plugins
↓
Warnings → select → install

→ Jenkins / Java-code-review - Config →
post-build Actions → Record Compiler warnings and static analysis results

tools → pmp
Checkstyle → save
findbugs

Build Now → Check for Warning

→ New item
Java-test 1 → save

→ Same Config for Config
only change is goals

goals → test → save → build Now

→ add built step → Execute shell → add

Note: cd /var/lib/Jenkins/~~data~~/Workspace
↳ ls ↳ cd . all proj will be stored
↳ sudo apt get install xsltproc

↳ cd target
sudo xsltproc -o ./surefire-reports
/index.html ../sample.xsl ./surfire
-reports/*.xml

→ Post Build Actions
→ Publish JUnit test result report
→ target/surefire-reports/*.xml
→ setting → plugins → publish → HTML Publisher → install

→ Java test 1 → Config → add post Build action
↓
Publish HTML Report
↓
add
↓
target / ~~sheet~~ surefire-reports
↓
index.html
↓
save

→ Build Now

to check →
Java test → Workspace → Check → install packages
→ Build step → Execute shell
cd target
java -jar *.jar

→ New project →
→ Java-package-deploy config →

← Use it here not above

to check →
Java test → Workspace → Check
execute shell
→ cd target

Install pipeline: to contain all the pipelines
→ Setting → plugin → ~~Add~~ Available plugins
→ Build pipeline → Install

→ Jenkins → (+) → Build pipeline View

↓

Java-ci-cd

↓

Create

↓

Java-Compile

↓

Save

→ Jenkins / java-Compile / Config → Post-build Actions

↓

Build other project

↓

Java-Code Review

↓

trigger only if build is stable

↓

Save

→ Do the same thing for all the project to form a pipeline as in picture

→ Run the pipeline once Build

Sir did this: we have to do

Note:

Tomcat / Glassfisch → webserver for Java [JSP → Java single Pages]

6th: Java JSP WebApp

→ Checking pom.xml
↳ maven required
↳ WAR pakageing or Ear
↓
Web App packaging

→ Jenkins / Jsp App deploy → Config → Source code Manager
↓
git repo
↓
Branch → Main
↓
Built steps → Invoke top-level Maven targets
↓
main - ~~version~~ maven (Change this)
↓
goals
Clean Compile Validate
-P metrics ~~nm~~ pmd : pmd
test install verify
package

post Built steps → Record Compiler warning and statistical report → find →

pluggins → deploy war/ear install
Post Build action → deploy war/ear to a Container
(Configure tomCat into diff {port 8081} or try this (Assignment)
↓ **/*.war

Context path → MyMavenApp

+ Add to container → tomcat 9.x Remote
↓
Credentials selected
↓
tomcat URL
http://172.18.182.36:8081
↓
save → Build

Click : Browser : 172.18.182:22:8081/
MyMavenApp

~~Setting~~
Settings → Manage Jenkins → Credentials → System → Global Credentials (unrest)
↓
add the one Created on tomcat

22/May/2026
~~sudo~~ Python demo:
~~Dev~~ Clone ~~git~~ sudo git clone https....
→ Create Virtual environment → Packages limited only to that particular ~~fold~~ folder

⇒ cd python-demo-app-monitoring

⇒ Where python3 --version

⇒ Sudo apt ~~get~~ install python3 ~~apt3~~ python3-pip

→ cd src

⇒ ls

⇒ sudo python3 -m venv myenv

⇒ source myenv/bin/activate

⇒ sudo apt-get install python3.12-venv

⇒ sudo python3 -m venv myenv

⇒ source myenv/bin/activate

⇒ pip3 install -r requirements.txt
(in case permition denied error)

↳ Change pip3 to sudo pip3 install -r requirements.txt

If it dosent work then change ownership, first check

→ ls -la (root user is there)

Change ownership

→ sudo chown -R msis:msis myenv/

⇒ ~~change pip3 to~~ pip3 install -r requirements.txt

→ ls -la

→ cd .. → check from file ⇒ makefile (python-demoapp-monitoring)

⇒ black --check src/ ∮ flake8 src/app/ ∮

flake8 src/run.py

⇒ black --check src/

⇒ pytest -v --junitxml=lint-results.xml

⇒ pytest -v

⇒ cd src

⇒ sudo python3 run.py

⇒ python3 run.py

Check → https://172.18.182.22:5000

$\underbrace{\text{://172.18.182.22}}_{\text{localhost:5000}}$

Check the python app

to automate this

⇒ Open jenkins

⇒ Create project → python Basement display

⇒ git link → Build steps → build shell → make lint

⇒ Build Now

⇒ Config → Execute shell

make lint
make lint-fix
make test
make test-report
$\boxed{\text{make run}}$

Replace :

fuser -k 5000/tcp ||
true

export BUILD_ID=
dontKillMe

nohup src/.venv/bin/python
src/run.py > app.log
2>&1 &

⇒ Build Now

Check localhost:500

Assignment → Make sure the run happens to execute it on a Background instead of foreground

⇒ deactivate

⇒ Install Java JDK → How to install apache tomcat
vultr.com ↑

⇒ sudo apt
follow this link

⇒ sudo groupadd tomcat

⇒ sudo useradd -s /bin/false -g tomcat
-d /opt/tomcat
tomcat ~~to~~

⇒ wget -O tomcat.tar.gz https://dlc - - - -
. . 11.0.15/. tar.gz

⇒ sudo rm

⇒ sudo nano /opt/tomcat/conf/server.xml

~~Reset~~ Personal ctl ~~root~~ -u tomcat.service

⇒ https://172.18.182.36:8081

---

framework testing
- Junit → Java
- pytest → Python
- Mocha → nodejs

8:) Node js Monitoring app:

↳ js front-end scripting Language

↳ ~~index~~ index.js, or Main.js ~~...~~ or Server.js

↳ Install dependencies → in java pom.xml

→ in JS → package.json

→ in python requirements.txt

→ mycrosoft → Sln file (solution file)

[pip, maven, apt, npm ~~...~~ Package manager

pip → python, maven → Java, apt → os, npm → js, nvm

Commands & steps: →

↳ sudo git clone https://. . . . .git

↳ cd nodejs-monitor-devops/

↳ ls

↳ sudo apt-get install nodejs npm

↳ node js --version

↳ npm -version

↳ cd src

↳ ls

(No node_module in src)

↳ Sudo npm install

→ npm audit fix --force
→ sudo npm ~~test~~ lint → sudo n
→ node server.
→ eslint . --ext mjs && prettier --check **/*.mjs
(Copied from package.json gitflsrc)

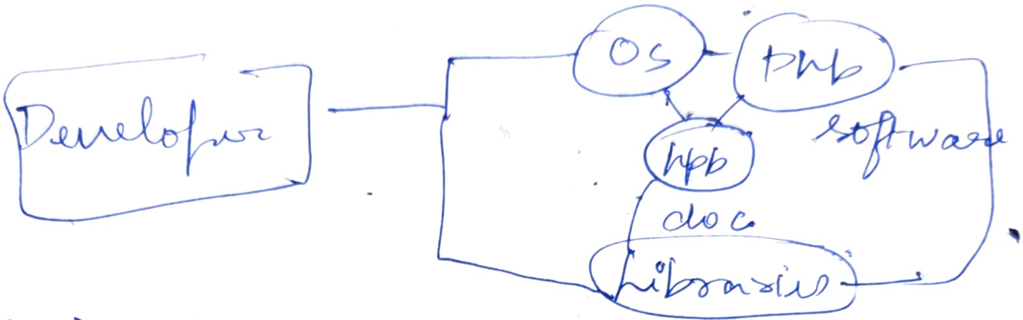
→ ~~sudo npm install~~
→ sudo npm start
(Goto Browser localhost:3000)

Now Lets try with Jenkins (automate task);

↳ Jenkins → node-js-basmtlla deploy
↳ Execute shell → make lint
make lint-fix
make test
make test-report } Not run
make run
↳ Build
↳ Sudo lsof

Docker:

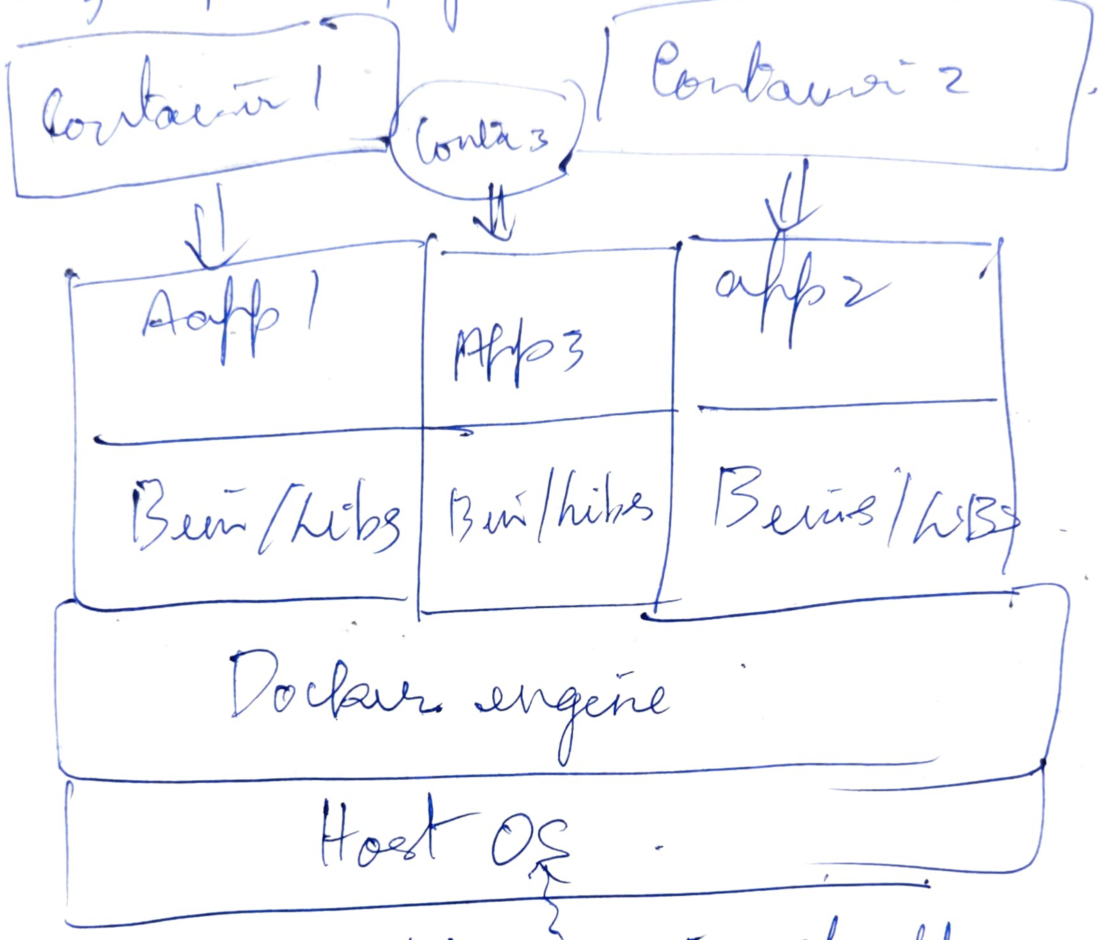
- Container [Software platform for Containerizing the application]



- No hardware or Software dependence During Deployment
  → Continerization is OS level Virtualization.
- Init the OS utilities or optimize OS
  - File system
  - tools
  - libs / Runtime libs.
  - Binaries.
- Eg:- JVM → Java Virtual machine (Packaging of ByteCode).
- Docker is a Computer program that performs OS Virtualization, also known as "Contanizaration". It was 1st release in 2013 and is developed by Docker, Inc. Docker is used to ~~run~~ run software packages called "Containers".

↳ Docker, Alexis, Rocket Container, ContainerD
(Comes with ecosystem of tools in comparision with other)

↳ Light Optimize Operating System



Root file system should match.

⊙ Windo Host → Linux, Redhat, Ubunu

$\underbrace{\text{Linux, Redhat, Ubunu}}_{\text{Possible}}$

No possible.

⊙ Windows ⇝ Comes with windows system, Linux system
Combined.

Virtual machine (→ them full OS): Hardware → Host OS → Hypervisor (Type I & Type II) → Guest OS → App

Hypervisor → Zen, Matrix

Docker Components:

↳ Docker Hub is Docker's very own cloud repository

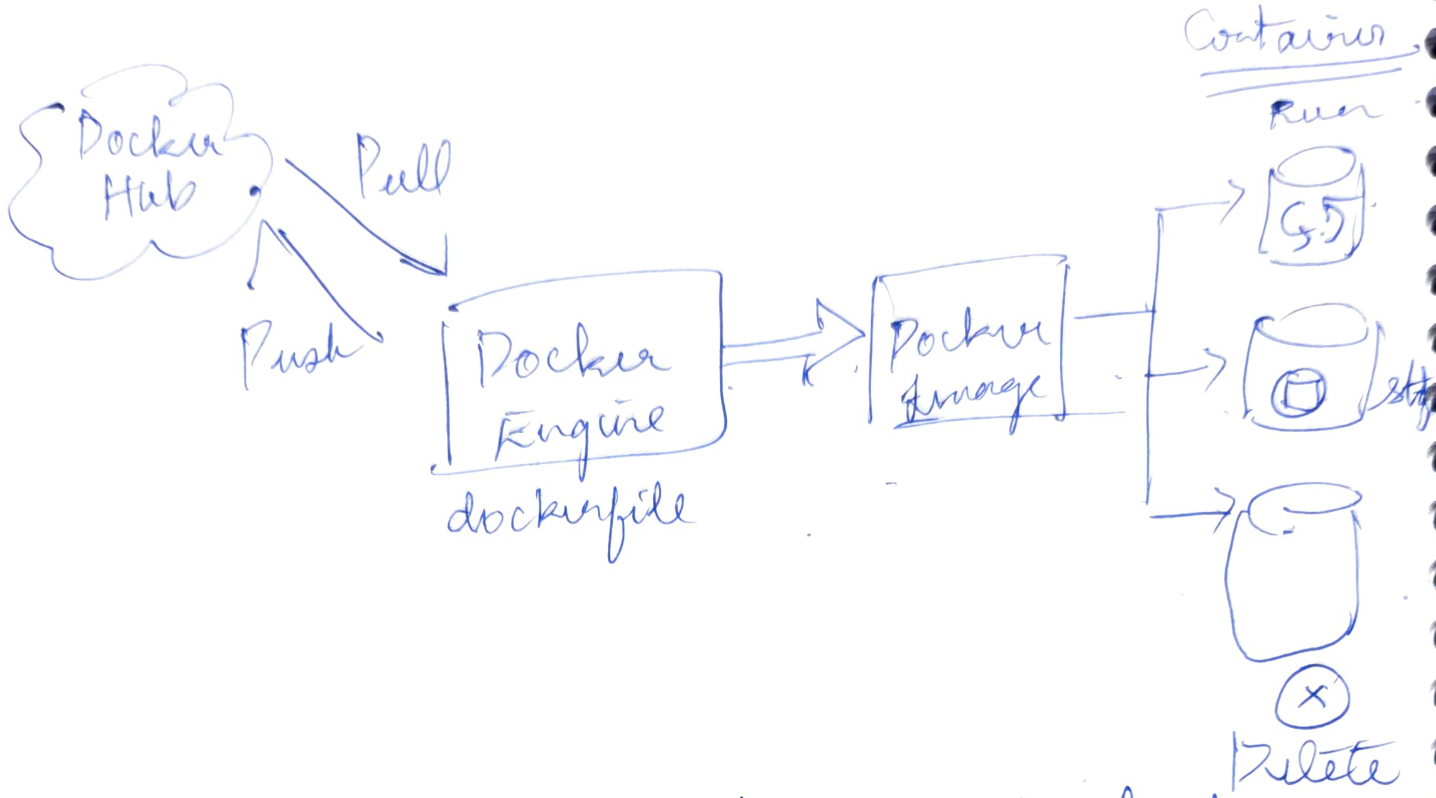
↳ Docker Registries:
↳ Control where your images are being stored.
↳ Integrate image storage with your in-house development workflow.

↳ Amazon → ECR
↳ Github → GSCR

↳ Storage space used to store Images.
↓
Readonly template that holds whatever needed to run the application (OS, Binary, Dependencies, libs & tools)

↳ Run Image → Container is Created

↳ ~~Image~~ → Dockerfile ⇒ Docker Image (Stored in Dockerhub registries)
⇓
Docker Containers.



All Containers run on root user, localhost Name.

- Install Docker engine
  - sudo apt-get install docker.io
  - Sudo docker pull centos:8
  - Sudo docker pull Ubuntu:latest
  - Sudo docker image ls
  - Sudo docker run -it -d --name Centos Centos:8
  - Sudo docker ps
  - Sudo docker ps -a
  - (to go inside Container)
    Sudo docker exec -it centos /bin/bash

↳# cat /etc/os-release → will tell us the version of OS.
↑
Inside Container

↳# Exit

⊙ to stop & remove Container:
↳ sudo docker ~~stop~~ rm -f Centos → for full Worst Case
or
↳ sudo docker stop Centos } this should
↳ Sudo docker rm Centos } be done.

Note:
Docker Demon → docker → Clm Comand
↳ It uses other runtime to pull, going & the execution using docker Demon & docker Clm

Manual method of Packaging: Not Recomended.

↳ sudo docker pull ubuntu:latest
↳ sudo docker run
↳ sudo docker run -it -d --name myubuntu ubuntu:latest
↳ sudo docker exec it myubuntu /bin/bash
↳# apt-get update → update apt first
↳# apt-get install apache2 nano
(Create dummy html in /var/www/html
↳# nano /var/www/html/test.html
(Paste html page)

↳# Cat /Var/www/html/test.html

~~↳# cd /var/~~ hash

↳# cd /Var/www/html/test.html.

↳# ls

↳# - rm -rf index.html.

↳# cp -f test.html index.html.

↳# exit

↳ sudo docker ps

(Container ID → 703daa95b78b of Ubuntu)

↳ (taking snapshot of container) ⇝ Image.

Sudo docker Commit 703daa.....

myapp:V1

Name given.

↳ (Verify if image is created or not)

sudo docker image ls.

↳ Sudo docker run -it -d --name

myapp -p 81:80

myapp:V1

(Port mapping to the host port to access to outside

localhost:81

↳ sudo docker ps

↳ sudo docker ps | grep myapp

searching in case of multiple containers running

↳ sudo docker exec -it myapp /bin/bash

↳# service apache2 status

↳# service apache2 start

↳ Exit

(It will have its own docker network IP address)

Note:

[Now check on chrome and say localhost:81

we get the web page.]

[Register in Docker hub repo]

hub.docker.com/repositories/shethalrao94

[Cafe static website ①]

Lets automate the process:

↳ cd Cafe static-Website

↳ ls

↳ sudo nano Dockerfile

↳ ls

↳ sudo docker build -t mycafeapp:v1 .

[In case the Dockerfile in some other location or named diff use -f specify

↳ sudo docker run -it -d --name Cafeapp -p 82:80 myCafeapp:V1

Check in localhost:82

↳

Now check going to some other lab pull it and Run it

① Tag an image (tag to User handle name ~~[illegible]~~)

↳ sudo docker tag myCafeapp:V1 shethalrao94/myCafeapp:V1

↳ sudo docker images

↳ sudo docker logout

↳ sudo docker login -u

↳ sudo docker login -u shethalrao94
→ enter the autentication Pin token

↳ sudo docker push Shethalrao94/myCafeapp:V1

Now check in docker hub repositoris

Now ~~do~~ check pulling the doc and check in other lab

→ cd

→ ls

→ cd node-js-monitor · -doup-/

→ ls

→ cd Build

→ nano Dockefile

→ DockerFile understand

Note: Alpine Light OS → Node Version 16

LABEL → Extra metadata that will be available at runtime

ENV → Environment Variable set

WORDIR → Create a folder Called app → root file system (Under the)

Dokerfile → Requirements of Image
- OS
- Software Packages

Packet.jason → Dev dependency / Prod dependency } Hence - production

→ Cd ..

→ Sudo docker build -f build/Dockerfile -t nodemonitoringapp:v1 .

(Building an image)

(Run Container)

→ Sudo docker run -it -d --name nodeapp
-p 3002:3000 nodemonitoringapp:v1

→ Check → localhost 3002.

[Lets try localhost 8080 → for Jenkins]

New project → Nodejs_docker_dep

↳ Copy git URL → master

↳ Build Step → ~~Execute shell~~ → Save

↳ setting → available → Docker build & publish → Install

CloudBees

↳ Nodejs-dock → Config → Docker Build and Publish

Note: Provide path and automatically build image.

Credentials of Doc hub.

Repo Name:

Shuthol 123/nodemonitoringapp tag: V1.

Dockerhub Credential.

(1st add Credential

Setting → Credential → add Credential)

Registry Cred: Shuthol

Advanced>

Build Context: .

Dockerfile path: ./Build/Dockerfile

[if Error :

→ Sudo docker ~~xxxx~~ chmod -R 7779

/var/run/docker.soc .

→ Sudo usermod -aG sudo jenkins

→ Sudo.

refer Screenshot and txt file] .

[artifact → Docker Image]

(Install all the Docker libs → in Picture)

→ add a build Step :

↳ Execute on Docker Command

Docker Command :

↳ Create Container

↳ Image No .

shuthalerao94 /nodemonitoringappv1

↳ Container name : nodemonitorapp

↳ Advanced

↳ Port Bindings : 3005 : 3000

↳ save

Open on Cmd → Sudo nano /lib/systemd /

system/ docker .

service

~~→ Sudo nano /~~

Copy the Comand after Commenting → Execstart = /usr/bin/dockerd -H . . . . . .

Exec

→ ~~sudo~~ save the changes

⇒ sudo systemctl daemon-reload
⇒ sudo systemctl " reexec
⇒ sudo systemctl restart docker.service

Jenkins → Manage Jenkins → System →

docker builder →
unix:///var/run/docker.sock

⇒ nodejs-docker-deploy → Execute Docker
→ Execute docker command
New add build step
↳ start container
↳ nodemonitorapp

localhost:3005

Building a Pipeline:

⇒ sudo docker ps
(already nodeMonitorapp is executing Hence not have to stop it)

⇒ sudo docker rm -f nodemonitorapp
→ sudo docker ps → (Cross check).

⇒ Jenkins → node-barometal-deploy

make lint
make lint-fix
sudo npm install pm2 -g

cd src && sudo pm2 start server.mjs

→ post build Actions
↳ trigger parameterized build on other -
↳ nodejs_docker_dev -

→ new View
↳ nodemonitor_ci_cd
↳ select initial job -
↳ nodejs_barental_deploy

save

or

Edit View

↳ node monitor ci-cd
~~up~~stream / downstream Config
nodejs -basemultle_deploy

If error:

→ sudo visudo
↳ edit the file
↳ # sudo
↓ add Below
jenkins ALL = (ALL) NOPASSWD: ALL

→ sudo systemctl restart jenkins
Restart Jenkins → Run Pipeline

10th one: Clone it

→ sudo git clone "10th file link".

→ cd docker-vol

→ sudo docker build -t nodevolumeapp:V1 .

→ Sudo docker run -it -d --name nodevolumeapp -p 85:80 dockerapp :V1

→ localhost:85

- title Customer1.text - Heyapp
- title Custom 2.txt -

→ Sudo docker ~~exit~~ -it nodevolumeapp /bin/bash

- ↳ ls
- ↳ cd feedback
- ↳ ls
- ↳ exit

→ Sudo docker rm -f node volumeapp

→ localhost:85 (wont be able to read)

→ Sudo docker run -it -d --name nodevolumeapp -p 85:80 nodevolumeapp:V1.

→ sudo docker exec -it nodevolumeapp /bin/bash

↳ cd feedback
↳ ls
↳ exit

⇒ sudo docker rm -f nodevolumeapp

Note:

Docker Volume → Mount a folder of the system to have info after deletion

3 diff (testing phase)
↳ anonymous Volume → -v
↳ Bind mount Volume (Host Mount) (Host & Container)
↳ Docker managed Volumes (Production Volume)

⇒ 1st ~~Create an anonymous Volume~~ : anonymous Volum

⇒ sudo docker run -it -d --name nodevolumeapp -v mynodevolume:/app/feedback -p 85:80 nodevolumeapp:v1

⇒ sudo docker exec -it nodevolumeapp bin/bash
↳ cd feedback
↳ ls
↳ Exit

⇒ sudo docker run -it -d --name nodevolumeapp -v mynodevolume:/app/feedback -p 85:80 nodevolumeapp:v1

=> Remove with -rm Container

-> 2nd type: (Bind Volume)

=> mkdir bindvolume

=> Sudo mkdir bindVolume

=> ls

Running in background
↳ demon mode

=> sudo docker run -it -d --name nodevolumeapp -v ./bindVolume:/app/feedback -p 85:80 nodevolume:V1

=> Sudo docker exec -it nodevolumeapp /bin/bash
=> ↳ cd feedback / localhost -> Customer3.txt
↳ ls ↳ exit

=> cd bindVolume

=> ls

=> Sudo touch Customer2.txt

=> Same thing

=> Sudo docker rm -f nodevolumeapp

=> 3rd type: (Docker Managed Volume)

=> cd ..

=> Sudo docker Volume Create mynodeVolume

→ Sudo docker Volume ls
↓
list the Volumes listed

→ Sudo docker run -it -d --name nodevolumeapp -v mynodevolume:/app/feedback 85:80 nodevolumeapp:V1

→ Localhost:85 ↳ Createfile
→ Sudo docker exec -it nodevolumeapp /bin/bash

↳ Exit

→ Sudo passwd root
↳ University
↳ University

→ su root
↳ University
(Now we will be as Root@main's User)
↳ cd → (home folder of Root)
↳ cd /var/lib/docker/Volumes/
↳ ls (Now mynodevolume must be available)
↳ cd mynodevolume
↳ ls
↳ ~~do~~ cd _data
↳ ls → this is where data will be there

Note:

Docker Compose → Simple Yaml file where we can tye all the diff container and run them togather

[Absent fill in] → class was not there.

(5th)

Docker Compose: (Copy from Edi's Repo (1.12.18.182.22) install steps .txt → Run the code)

⇒ Check file docker-compose

⇒ Sudo curl -L "https://github.com/ ..... .compose-

⇒ 5th github Repo (Clone it if not done)

⇒ Cd Cafe-Dynamic-Website

↳ ls

↳ sudo nano docker-compose.yml
[write this file to define how service look like]

Note: Connection string → has database User name, Pass etc etc

Yaml → Declarative language

↳ cd mompopcafe

↳ ls

↳ sudo nano Dockerfile

↳ Expose 80

↳ Cd..

↳ sudo nano docker-compose.yml

Note: 2 Containers

↳ Cafeapp ↳ mysql

Note:
Env:
password → Ms018@123
Database :→ mom-pop-db
mysql-pass :→ Ms018@123
Port → 3307:3306
Volume:
- mysql-data:/var/lib/mysql ← stored under path ie Databases

↳ sudo docker-compose up --build -d

Used Not Req in case Want to Run in Background

↳ Check localhost:5000

↳ cd ..
↳ Sudo docker ps
↳ sudo docker exec ~~-it~~ -it mysql /bin/bash
↳ mysql -u root -p
↳ pass: Ms018@123
↳ show databases;
↳ use mom-pop-db
↳ show tables;
↳ exit
↳ cd /var/lib/mysql
↳ ls
↳ cd mop-pop-db/
↳ ls

What is ibd, frm files?

↳ ~~cd~~ cd / ⟶ Root User

↳ ~~ls~~ cd docker-entrypoint-initdb.d

↳ ls

[Image: docker it look at this folder and source it to the database]

↳ exit

Assignment:

[Try ④ php Ecommerce Dynamic Web App]

↳ and git clone " "

↳ ls

↳ cd php-mysql-app

↳ sudo nano docker-compose.yml

↳ control C to stop all containers

[↳ sudo docker compose up --build
↳ sudo docker-compose ~~up~~ docker-compose-update
↳ or → sudo docker compose down up]

port:
- 8008:80
both same port

Now Lets Do this with Jenkins:

Note: Multiple file docker diployment:

· localhost:8080

↳ Login

↳ New → Cafeapp-docker-comp-deploy

↳ git → link (5th)

↳ main

↳ Built step
☑ use Docker compose file.
↳ Docker Compose built step

↳ start all services (Docker Compose file

↳ Build

↑

Before doing all of this:

↳ cd ..

↳ cd Cafe_Dynamic_Website/

↳ sudo docker-compose down

Check:

localhost:5000

Exam: → Source Code in git Cumb

↳ Understand

↳ Jenkins project which should be able to Run and pull Repo from git

↳ Jenkins 2nd project to ~~next~~ automate run and localhost update → php/Java/Python.

↳ Jenkin 3rd project to run Kubernetes

Nodeport → Makes your application reach through internet with port Number → Bind to IP address of one of your node.

↳ cluster IP (for deployment)

↳ create a pod.

Kubernettis:

Create a Kubernettis Cluster

↳ Master node Kubernetis cluster

↳ Log in → minikube start

↳ Kubectl get nodes

↳ kubectl get nodes -o wide
(Make a note of IP address of the ~~server~~).
↳ 192.168.49.2 : 32467

↳ Clear

↳ kubectl get pods --all-namespaces

↳ Kubectl Create deployment ~~nego~~ nginx
--image=ngix:latest
deployment.apps/ngix Created

↳ Kubectl get deploy
↳ kubectl ~~get~~ describe deployment nginx.

↳ Kubectl get pods

↳ Kubectl expose deployment nginx --type=NodePort
--port=80 service/nginx exposed

↳ Kubectl get svc

↳ 192.168.49.2 :32467

↳ Kubectl Create deployment Cafeapp
--image=sreedocker123/cafewebstatic:V1

--replicas=3

(Check hub.docker.com)

↓

shuthalrao94/mycafeapp

↳ Kubectle get deploy cafeapp.

↳ Kubectl get pods.

↳ Kubectl scale deployment cafeapp --replicas=5

↳ Kubectl delete pod cafeapp-5fc5492864-6fslv.

↳ Kubectl ~~...~~ expose deployment cafeapp --type=LoadBalancer --port=80 --target-port=80

service/cafeapp exposed

↳ Kubect get SVC

↳ Kubectl set image deployment cafeapp cafewebstatic=shuthalrao94/mycafeapp:v2

↳ Kubectl get pods.

↳ 192.168.49.2:32467

~~↳ Kubectl~~ ..

↳ kubectl get ~~deploy~~ deploy
↳ kubectl get svc
↳ kubectl delete deployment nginx
↳ kubectl delete deployment cafeapp
↳ kubectl get pods
↳ kubectl delete svc nginx cafeapp

① Copy ① from sirs Repo
↳ sudo git clone ______
↳ Cd to Cafe static
↳ nano cafe-static-deployment.yaml
↳ nano rolling-updated-deployment.yaml
↳ nano Cafestatic-service.yaml
↳ kubectl apply -f Cafestatic-deployment.yaml
↳ kubectl get pods
↳ kubectl get deploy
↳ kubectl get pods -o wide
↳ kubectl apply -f Cafestatic-service.yaml
↳ kubectl get svc
~~exec~~ nginx link // ______ port No (Nodeport

CICD Pipline using Kubernetes:

↳ Down Developing tools → Kubernetes

↳ Jenkins.

↳ Settings → pluggains → advance setting → .

~~Plugging~~ → Browse

↳ plugging → Kub → Kubernetes

Kubernetes CLI

Kubernetes Credential

Kubernetics Client API .

↳ add Credential → click on Cridential

↓

Kube Configuration

add Cridential bla bla. blaa blaa - - - -

↳ ~~terminal~~ gedit ~/. Kube/Config

↳ ~~gedit~~ Cat ~/. Kube/Config

↳ ~~sudo apt-get install~~

↳ coppy entire Content

puth in under Kub Cred .

global

Kube-Cred-83

Kube-Cred-83 .

Kube Config

~~Select~~ ↳ Create a freestyle project in Jenkins

↳ go to directory listing

↳ ~~Nodejs Monsoon~~ Cafe staticweb

↳ Note whats deployment.yaml

↳ Cafeapp-blue-deploy

↳ freestyle

↳ git — URL

↳ */main

↳ Add Build set

↳ Deploy to Kubernetes

↳ Configfiles

↳ Cafestatic-deployment.yaml, Cafestatic-service.yaml

↳ save

↳ Build

Lets deploy ③ one

↳ bookalbum-deploy → git → URL

↳ Master

↳ Build Step

↳ Deploy to Kubernetes

↳ Kube Config

↳ Kube Cred-83

↳ Bookalbum-deployment.yaml,
bookalbum_service.yaml

Build

↳ Kubectl get deploy

↳ Kubectl get pods

↳ ~~Kubectl get dc~~

Build CI CD

Pull → test → Create Docker image push
↓ ~~Create~~ pull image and
↓ then Execute the Application.

↳ Clone Cafe Dynamic Web.
↳ cd Cafe-Dynamic.
↳ Cd deploy.
↳ Cd Kubernetes-Manifests/
↳ ls
↳ nano mysql-pass.yaml.

Note:

↳ nano Cafe-app-Config-map.yaml -
↳ nano mysql-pv.yaml.
↳ nano mysql-pvc.yaml
↳ ~~Ssh mode@172.18.182.~~
↳ Kubectl get node.
↳ cd Cafe-DynamicWebsite
↳ cd Deploy
↳ Cd Kubernetes-Manifest
↳ Kubectl get pods
↳ Kubectl apply -f mysql-pass.yaml.
↳ Kubectl get secrete or discribe Secrets.
↳ Kubectl apply -f mysql-pv.yaml

↳ kubectl get pv

↳ kubectl apply -f mysql-pvc.yaml

↳ kubectl get pv

↳ kubectl get pvc

↳ kubectl apply -f Cafe-app-config-map.yaml

~~↳ 1st do next dep.~~

↳ kubectl apply -f Cafe-app-web-deploy.yaml

↳ " " " " web-service.yaml

↳ kubectl get pods

↳ kube - - - - , , Cafe-app-db-deployment.yaml

db-service.yaml

↳ kubectl get svc

↳ kubectl get configmap

↳ kubectl edit configmap app-config

i → for editing → change ip to "mysql"

Exit-ship - ; :wq ↵

↳ kubectl rollout restart deployment cafeweb

Building CI/CD pipe:

↳ Dev Docker
↳ Prod Kubernetes (QA or Staging env)
↳ Nodejs Monitoring app (8) → listing
↳ Add Project on Nodejs (Jenkins):
↳ nodemonitorapp-kubernetes-deploy
    ↳ Deploy to Kubernetes
        KubeConfig
        ↳ Kube-Cred-83
        ↳ Config files
            ↳ Nodeapp-deploy.yaml,
                nodeapp....yaml

Nodemonitor CI-CD
↳ Config →
    Execute shell
    make lint
    make lint-fix
sudo npm install pm2 -g
cd src && sudo pm2 ~~start~~ stop server.mjs &&
sudo pm2 start server.mjs

trigger (....)
↳ Docker Project
    Docker build & push

↳ Explore post Build
30005 : 3000

Execute Docker Command
↳ start Container()

↳ post Build
Projects to Build
↳ Kubernetes Deploy

Node's Developmental-deploy → Node's docker-Deploy-bda | nodemonitorapp-Kubernetes.

Errors debugging:

→ sudo docker ps
→ Sudo docker rm -f Nodemonitorapp
→ Sudo lsof -i 3000
→ Kube delete deploy nodeapp
→ Kubectl get svc

localhost :3000
Localhost : 3005

Nodeapp service
Cluster IP
172.... :30100

→ Kubectl

8080

→ Groovy Script or Pipeline Script
(Reuse the Same Script of Multiple project
& Reuse them for Multiple project)

↳ Control A → Ctl + C

↓

go to jenkins

⇓

Create a project

⇓

New item → pipline - project

⇓

pipline Script

Note : Change the Docker Credential id
and paste the pipline Script

Pluggin → Blue ocean

**/*.war

26/Feb

↳ Machine Learning - Deployment.

↳ 15th Machine Learning Workload Deployment.

↳ git Clone Repo - - - - -

↳ Free Style project → ml_deployment.

↳ git → https - - - - -

↳ Main.

↳ Build steps

↳ Execute shell

↳ #sudo python3 -m venv venv
~~python3~~ ~~. source~~ . venv/bin/activate
sudo pip3 install -r requirements.txt
sudo python3 ml_train.py
sudo python3 flask_web_api.py

--break-
system-
packages

3/3/2026 Ansible Configure Management :

(14) → Ansible PlayBook + slave / Nodes
↳ git Repo

↳ Clone the URL → git clone https....
↳ ls
↳ cd AnsibleWork Updated
↳ ls
↳ cd ansible-apache
↳ Sudo nano hosts
↳ Change the IP address that we need to Run the Node in and name of peers ip and yours ip
(insis-cd → (Mine ip insis)

38
36.

↳ sudo nano apache.yml
Change the name (insis).
↳ ~~Else~~
↳ Creating key pair : (do it only once)
↳ sudo ssh-keygen -t rsa
↳ Note : → (root/.ssh/id_rsa) : myansiblekey
↑ hidden file
passphrase : ↵
↳ ls

↳ sudo ssh-copy-id -i myansiblekey.pub
~~ ~~ users@172.18.182.36
host Node

↳ Sudo aptget install ansible ~~sshpass~~
sshpass

↳ Sudo ansible-playbook apache.yml -i hosts
-c ssh -K --ask-pass

↳ cd

cd -

~~Note: it~~ should show connected

Go to Browser → check → 172.18.182.38.

Next → Application Deployment

cd → AnsibleWorkUpdated → lampansible → ls

sudo cp ../ansible-apache/hosts .

ls

Sudo nano hosts → [lampserver]

↳ sudo nano . lamp.yml

Note: state: ABsent → Deletion or purge.

state: Present → Start

↳ idempotent → If the file exists then does not change else makes the changes.

↳ Note: Copy → which location & where Change /home/msis/Ansible Work/Updated /lampansible /~~lampstack~~ file......

↳ pwd

↳ sudo ansible-playbook lamp.yml -i hosts -C ssh -K --ask-pass ↵

↳ Refresh localhost:36 it should work

Installing Lampstack to Node Server:

↳ sudo nano lamp-app.yml

↳ php7.2 → php
php7.2 → php

src → home/msis/Ansible Updated/ lampansible/multipurpose

↳ sudo ansible-playbook lamp-app.yml -i hosts -C ssh -K --ask-pass

↳ Now ~~goto~~ ping on 172.18.182.36

Lab Exam:

$\boxed{\text{git}}$ → Baremetal → Containerization → $\boxed{\text{Docker}}$

↓

$\boxed{\text{Mini Kube Kubernetes Cluster}}$